

🎓 Education

University of Pennsylvania

Sep 2025 - Aug 2026

Visiting PhD Student

GRASP Lab · Advisor: Prof. Kostas Daniilidis

ShanghaiTech University

Sep 2021 - Jun 2027 (expected)

PhD Student in Computer Science

Research interests: Visual Localization · 3D Reconstruction · Event-based Vision

Mobile Perception Lab · Advisor: Prof. Laurent Kneip

ShanghaiTech University

Sep 2017 - Jun 2021

B.Eng. in Computer Science

🏢 Research & Project Experience

| Multi-view Geometry

A Linear N-Point Solver for Structure and Motion from Asynchronous Tracks

Oct 2025

ICCV 2025 [Highlight](#) · First author

- Extended two-view SfM to asynchronous timestamps and multi-view observations with a linear N-point solver covering global-shutter, rolling-shutter, and event cameras.

An N-Point Linear Solver for Line and Motion Estimation with Event Cameras

Jun 2024

CVPR 2024 [Oral](#)

- Developed a general linear N-point solver building on the 5-point model, improving numerical stability and increasing computational speed by more than 600 times.

A 5-point minimal solver for event camera relative motion estimation

Oct 2023

ICCV 2023 [Oral](#) · Co-first author

- Established a motion-geometry model for event-camera line features and designed a 5-point minimal solver to estimate camera translation velocity with known rotation.

Event-IMU velocity estimation

Mar 2026 - Present

- Fused event-camera and IMU observations for velocity estimation to initialize Event-IMU visual odometry.

| Marker-based Localization & Visible-Light Communication

Motion-Aware Optical Camera Communication with Event Cameras

Nov 2024

IEEE Robotics and Automation Letters, Presented at ICRA 2025 · First author

- Designed dynamic visual markers for event cameras and built a complete screen-to-camera visible-light communication system supporting localization and data transmission under camera motion.

Event-camera and LED-array localization and visible-light communication system

2024 - Present

- Built an event-camera and LED-array localization and visible-light communication system that improved EventOCC's runtime efficiency and operating range, and completed a working demo.

| Learning-based Feature Matching

Match-Any-Events: Zero-Shot Motion-Robust Feature Matching Across Wide Baselines for Event Cameras

Jun 2026

ECCV 2026

- Proposed a Transformer-based detector-free event matching network that extends local tracking to wide-baseline matching and enables zero-shot generalization across datasets.

| Other Research

Denoisied internal models: a brain-inspired autoencoder against adversarial attacks

Oct 2022

Machine Intelligence Research

- Designed a generative-autoencoder approach for class representation learning by jointly modeling denoising and class information to improve adversarial robustness.

 Experience

Changan Automobile (Chongqing, China) Aug 2021

Intern at Autonomous Driving Group

➤ Read internal documentation on autonomous-driving solutions and test scenarios, and attended discussions on underground-parking scenarios, user needs, and driver-monitoring anomalies.

ShanghaiTech University 2020 - 2022

Teaching Assistant

Machine Learning; Introduction to Programming; Computer Architecture

 Honors and Awards

Outstanding Student (Top 5%) 2022, 2024, 2025

Merit Student (Top 10%) 2023

Outstanding Thesis (Top 5%) Jul 2021

Second Prize, Shanghai Division, China Undergraduate Mathematical Contest in Modeling 2020

First Prize, Provincial Division, National Olympiad in Informatics Nov 2016

 Skills

Languages: C++, Python

Tools & Frameworks: PyTorch, ROS, NumPy, Eigen, Ceres Solver, CMake, Git, COLMAP, Pixi